

Test Anxiety Amongst College Students

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Background

In the realm of human experience, anxiety emerges as a complex interplay of psychological and physiological states marked by heightened apprehension and excessive worry, often preceding impending events. This ubiquitous sentiment touches every individual at some point in their lives, manifesting in various contexts and intensities. Notably, in recent years, college students have reported heightened levels of anxiety, attributed to a myriad of factors, including the challenges of adapting to new environments. This struggle becomes particularly pronounced in the context of a global pandemic, where every aspect of life, from education to daily routines, undergoes significant transformations. Amidst these changes, this research study seeks to look into a specific aspect – how the transition from remote to in-person instruction affected test anxiety levels among college students at UNC Chapel Hill?

According to a survey conducted during the 2021-22 academic year, involving 96,000 U.S. students across 133 campuses, 37% of them reported experiencing anxiety disorders (Shine, 2023). The pandemic ushered in numerous changes, and while some were positive, many posed challenges as individuals grappled with adjustments to new environments and lifestyles. In a survey conducted by Active Minds in April 2020, 80% of college students nationwide indicated a negative impact on their mental health due to COVID-19. Additionally, 20% reported a significant deterioration in their mental well-being (Horn, 2020). For students, this included the transition from remote to in-person learning. Those who already began college, understanding the norms, suddenly found themselves adapting to remote learning and testing. This forced a cycle of learning, unlearning, and relearning numerous concepts within a year as they shifted back to in-person testing. Understanding the impact of these shifts on students' mental well-being is crucial, especially in the absence of extensive precedent evidence to navigate such changes. This exploration serves as preparation for future scenarios where students might encounter shifts in the mode of learning. It holds the potential to reshape our perspectives on education, guiding institutions toward an evolved approach that prioritizes both learning and students' mental health.

Methods

For a comprehensive exploration of the impact of different modes of instruction on test anxiety among students, this study adopted a quantitative research approach. The sample pool comprised 44 participants selected from the University of North Carolina at Chapel Hill, with a deliberate focus on graduate students or recent graduates within the last two years. This strategic sampling aimed to capture nuanced insights from

individuals who navigated both online learning during the Covid-19 pandemic and the subsequent return to in-person instruction.

Recruitment involved administering a survey to participants, ensuring anonymity in their responses to a survey consisting of ten short questions. The primary independent variable, mode of instruction, was categorized into two distinct groups: remote and in-person. The study observed the impact of these instructional modes on various dependent variables, with emphasis on test anxiety. Additionally, the study delved into the exploration of how the mode of instruction influenced the GPA of students during both remote learning and the return to in-person instruction. This approach seemed to be the most accessible method to assess whether students' test anxiety translated into tangible effects on their academic performance.

This study employed a diverse array of statistical tools, including univariate, bivariate, and multivariate regression analyses. A bivariate regression was conducted to observe how GPAs changed during remote learning versus in-person classes. Survey questions, falling under the category of ordinal data, were analyzed using mode values and chi-squared tests to discern statistical significance. In addition, the study integrated a multivariate regression analysis, examining the influence of collected learning preferences on test performance, as reflected in GPAs. This methodological blend allowed for a comprehensive exploration of the multifaceted relationships within the data.

Results

Beginning with the impact of the mode of instruction on test performance (Q2), our findings revealed a significant association ($X^2 = 15.364$, $df = 1$, $p\text{-value} = 8.868e-05$). When comparing the p -value to a 5% significance interval, students' beliefs regarding how the instructional mode influences their test performance significantly differ between remote and in-person settings. However, when looking at anxiety levels during remote exams (Q3), our chi-squared test unveiled a moderate association between anxiety levels and response categories ($X^2 = 7.0909$, $df = 3$, $p\text{-value} = 0.06906$). Although falling short of the 5% cut-off, this finding shows a slight correlation between reported anxiety levels and specified categories during remote exams. Intriguingly, when investigating changes in test anxiety after returning to in-person testing (Q4), our analysis showed no significant association with response categories ($X^2 = 4.1818$, $df = 4$, $p\text{-value} = 0.382$). This implies that, according to the responses, the changes in test anxiety did not significantly vary among the specified categories after the shift back to in-person testing. This finding prompts us to ponder the factors that contribute to the stability of test anxiety amidst shifts in instructional methods. It's particularly noteworthy, given the relatively minor increase in anxiety levels observed during

remote learning among the sampled population.

As we explore the emotional facet of our participants' feelings about returning to in-person exams (Q5), we derived a median value of 5.5. This suggests that, based on the participants' responses to the question, the central tendency of their feelings about returning to in-person college exams falls at the midpoint of the scale. On a scale of 1 to 10, where 1 represents minimal anxiety, 5 is moderate anxiety, and 10 signifies intense anxiety, a median of 5.5 indicates a balanced distribution around the moderate anxiety level.

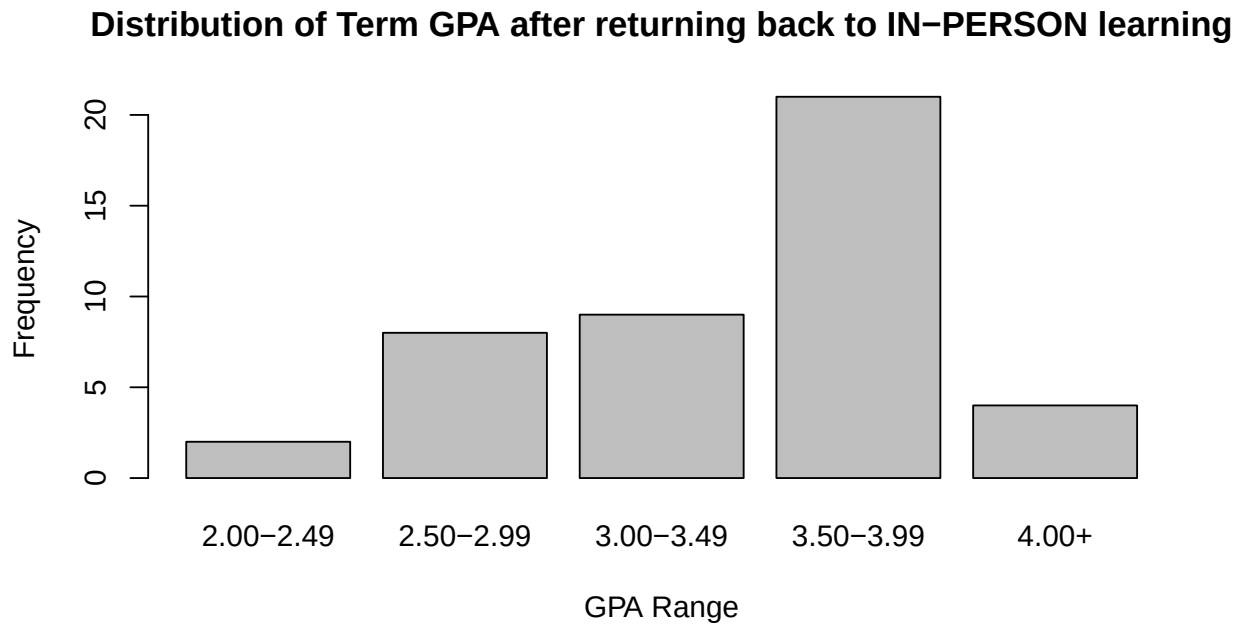
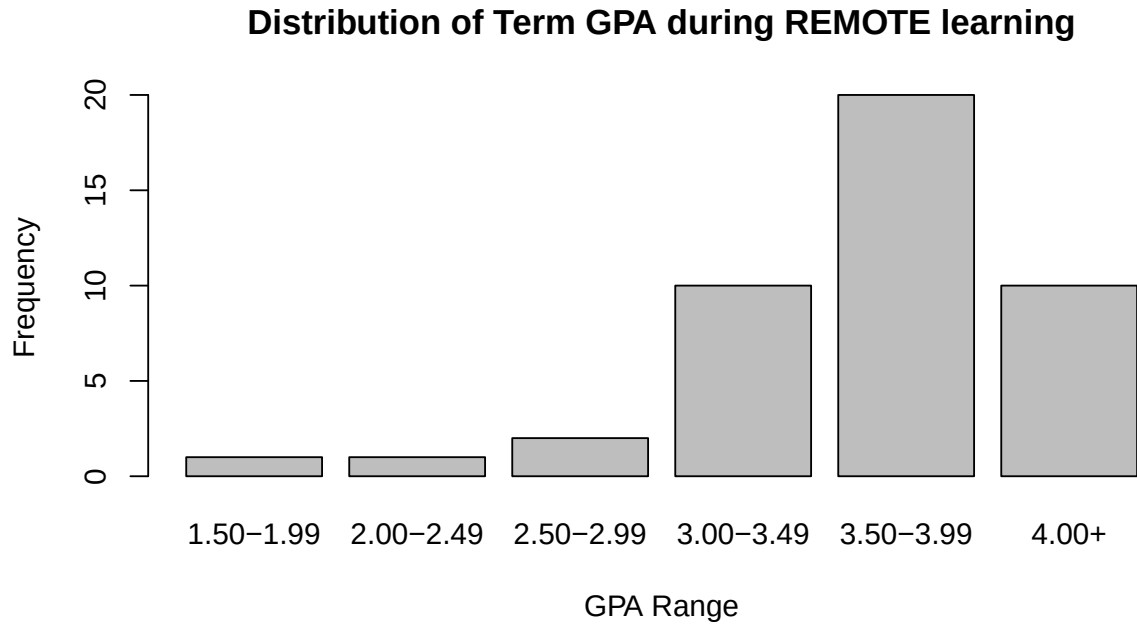
Exploring academic performance, my analysis of how remote learning impacts test performance (Q6) showed a notable association with response categories (X-squared = 16.909, $df = 4$, $p\text{-value} = 0.002013$). This suggests that participants' perceptions of the impact of remote learning on their academic performance vary significantly across different response options. When deriving the mode, most respondents agreed switching to remote learning had no effect on their test performance. This is particularly interesting because in contrast, the influence of in-person learning post-pandemic on test performance (Q7) showed no significant association with response categories (X-squared = 4.1818, $df = 4$, $p\text{-value} = 0.382$). However, most respondents concluded that switching back to in-person learning "somewhat improved" their test performance.

An intriguing finding is that respondents consistently reported lower test performance during remote learning, yet a majority expressed a preference for this mode, particularly in terms of alleviating exam-related anxiety. This preference may be linked to the nuances of course structure and exam administration. The statistical analysis for Q8, a chi-squared test, yielded a $p\text{-value}$ of 0.01163. This result indicates that students perceive notable distinctions in the efficacy of mitigating test anxiety between the specified instructional modes.

The linear regression analysis examining the relationship between students' anxiety levels (Q5) and their perceptions of instructional modes (Q8) revealed interesting insights. The coefficient for "No significant difference" (-1.6458) suggests that, on average, students who did not perceive a significant difference in the effectiveness of instructional modes experienced a decrease in anxiety levels compared to those who reported a difference, although this difference was not statistically significant. On the other hand, the coefficient for "Remote" (0.3693) indicates a minimal positive effect on anxiety levels for students who perceived remote learning as more effective. However, the $p\text{-value}$, 0.2267, suggests caution in generalizing these findings to a larger population due to it being significantly higher than the significance interval of 5%.

In concluding the study, the distribution of term GPA during remote learning (Q9) and post-return to in-person learning (Q10) offers a holistic view of academic performance. While both datasets initially exhibited a range of 3.50-3.99 for the mode, closer examination through frequency tables reveals a notable shift in the overall distribution. After the return to in-person learning, student GPAs became heavily right-skewed,

indicating a substantial impact on those affected by the change in instructional mode. The dataset now shows a higher concentration of GPAs in the 2.00 to 3.49 range, with a notably reduced number of 4.00s. An interesting outlier within the 1.50-1.99 range demonstrated improvement as they progressed to the next level since returning to in-person instruction.



A linear regression model was used to show the association between the mode of instruction and academic

performance during remote learning, as measured by the term GPA (Q9_numeric). To enhance the model's efficiency, GPA ranges were streamlined into more manageable categories; respondents in the 3.00-3.49 were now categorized with a GPA of 3.00. The model revealed that when students perceived remote learning as the mode most effective in alleviating test anxiety (Q8), there was a statistically significant positive effect on their term GPA. The coefficient for Q8Remote (0.4205) indicates that, on average, students who reported remote learning as beneficial for test anxiety experienced an increase in term GPA compared to those who did not find a significant difference. However, it's noteworthy that the variable indicating no significant difference in test anxiety alleviation did not achieve statistical significance. The model's overall fit is moderate, explaining approximately 13.13% of the variability in term GPA. While the p-value for the F-statistic is slightly above the conventional threshold, suggesting caution in generalizing these findings, the results imply a noteworthy connection between students' perceptions of remote learning's impact on test anxiety and subsequent academic performance during this instructional mode.

Examining the relationship between instructional modes and academic performance post-return to in-person learning, as indicated by term GPA (Q10_numeric), another linear regression model was used. The residual values, representing the difference between predicted and actual GPAs, ranged from -1.1875 to 0.8182, suggesting a reasonable fit of the model to the data. The intercept and the variable denoting no significant difference in test anxiety alleviation held statistical significance, however, the variable for remote learning did not. The residual standard error, measuring the average difference between observed and predicted values, was 0.5316, indicating a moderate level of accuracy in predicting GPAs. The model's p-value for the F-statistic is high (0.9606), emphasizing the weak impact of the model in predicting term GPA based on perceptions of instructional modes.

Conclusion

The study delved into the intricate relationship between instructional modes, test anxiety, and academic performance. However, it's crucial to recognize the constraints, such as potential issues related to generalizability due to the specific sample size and demographics of the University of North Carolina students. Remote learning, while associated with lower test performance, exhibited a positive correlation with alleviating test anxiety. This implies that, amidst academic challenges, students lean towards remote learning for its potential to ease exam-related stress. However, multiple inconsistencies surface, indicating respondents' contradicting perspectives. This discrepancy suggests that the topic might have been given insufficient consideration, as respondents express divergent views on the impact of the mode of instruction on test anxiety.

Exploring whether test performance and test anxiety are mutually exclusive would be the ideal avenue for future research, potentially giving an answer to the contradictions of this study.

These findings bear significant implications for various stakeholders involved in educational policy and decision-making. University administrators could benefit from understanding the preferences of students, particularly when it comes to instructional modes. Designing flexible and adaptive educational strategies that consider students' anxiety levels may contribute to a more supportive learning environment. Educational policymakers at the regional or federal level can draw insights from these results to inform broader strategies for responding to similar challenges in other institutions or during future crises. By recognizing the preferences and concerns of students, policymakers can tailor interventions and guidelines that align with the needs of diverse student populations. More importantly, exploring this topic can encourage students to prioritize their mental well-being and gain insights into the instructional modes that suit their needs. This, in turn, equips them not only for academic success but also fosters mental preparedness to navigate unforeseen challenges in the future. In conclusion, while the study provides valuable insights, it is crucial to acknowledge its limitations. Future research should aim for larger and more diverse samples to enhance the generalizability of findings.

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